• OS homework 1st

Introduction

Which of the following instructions should be privileged? ACEFGH

A) Set value of timer. B) Read the clock. C) Clear memory. D) Issue a trap instruction. E)Turn off interrupts. F) Modify entries in device-status table. G) Switch from user to kernel mode. H) Access I/O device.

OS structures

1. What is the purpose of system calls?

2. What is the main advantages of the layered approach to system design?

3. What are the advantages and disadvantages of the microkernel approach?

Process Management

1. What are the advantages of multiprogramming?

2. In operating system, each process has its own _____

   a) open files

   b) pending alarms, signals, and signal handlers

   c) address space and global variables

   d) all of the mentioned

3. The portion of the process scheduler in an operating system that dispatches processes is concerned with _____

   a) assigning ready processes to waiting queue

   b) assigning running processes to blocked queue

   c) assigning ready processes to CPU

   d) all of the mentioned

I、ACEFGH

II、1. ① Provide the interface between user programs and the OS.

② Allow programs to request OS services.

2. ① Modularity: Easier debugging and system verification.

② Simplicity: each layer uses only the services of the layer below it.

③ Maintainability: changes in one layer don't affect others much.

3. Advantages:

① Better reliability and security

② Easier to extend and port

Disadvantage:

Performance overhead due to more context switches and message passing.

III、1. ① Increase CPU utilization, reduce idle time.

② Increase throughput: Do more tasks in a shotter time.

③ Improve resource utilization.

2. d.

3. C.